LAB # 04

INTRODUCTION TO XILINX ISE AND Spartan 6 BOARD

IMPLEMENTING MUX WITH SPARTAN 6

Spring 2024

CSE-308L Digital System Design Lab

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Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Engr. Shahzada Fahim Jan

March 24, 2024

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Objectives of The Lab:

- Implementing 2x1 and 4x1 MUX on Spartan 6 board with dip switches and leds

Software Used:

- Xilinx ISE

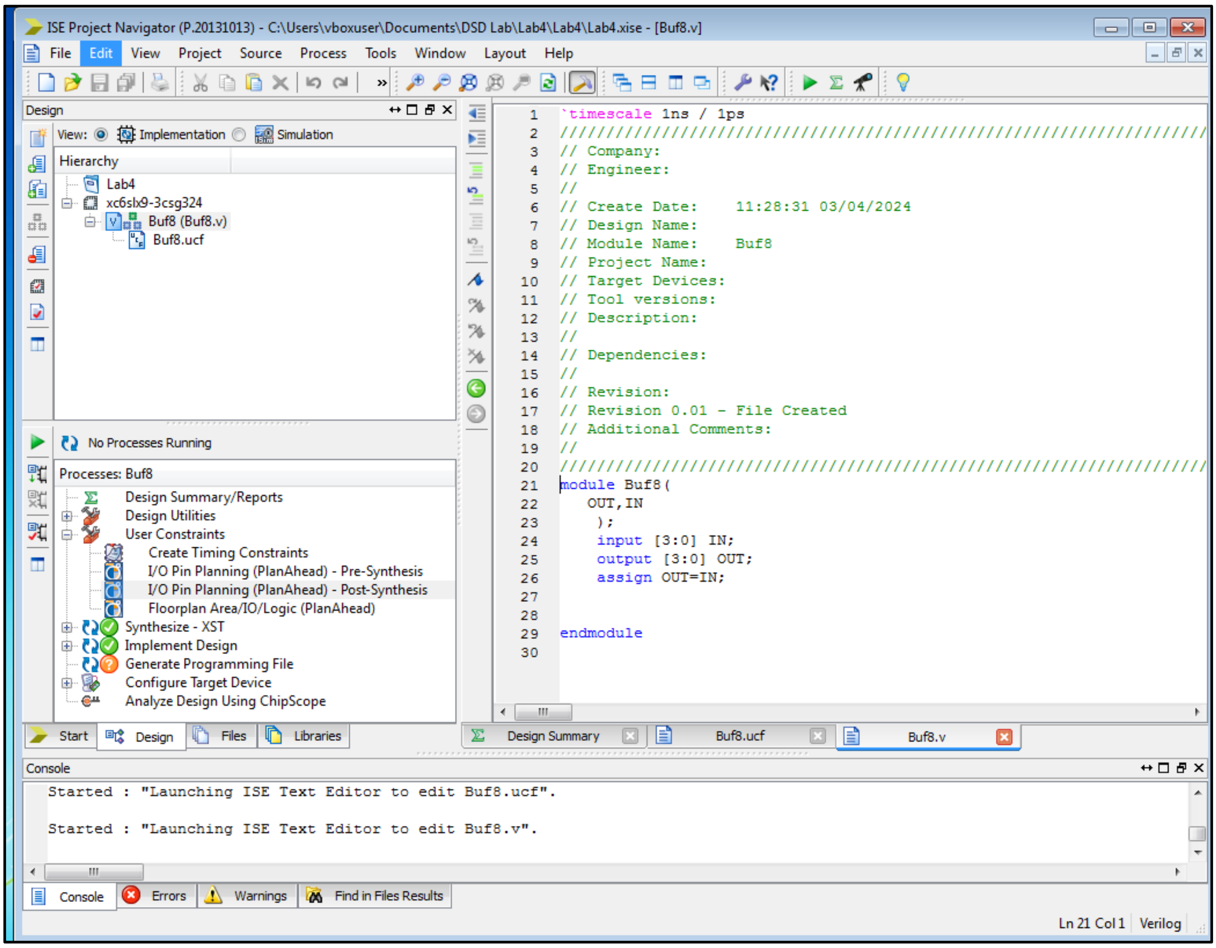
Lab Tasks:

1. Develop a program to control on Board LED using on-board available switch.
2. Develop a program that implements a 2x1 multiplexer on the board. Connect the inputs to switches and output to led.
3. Develop a program that implements a 4x1 multiplexer on the board. Connect the inputs to switches and output to led.

Task 1:

Develop a program to control on Board LED using on-board available switch.

Xilinx ISE Software:



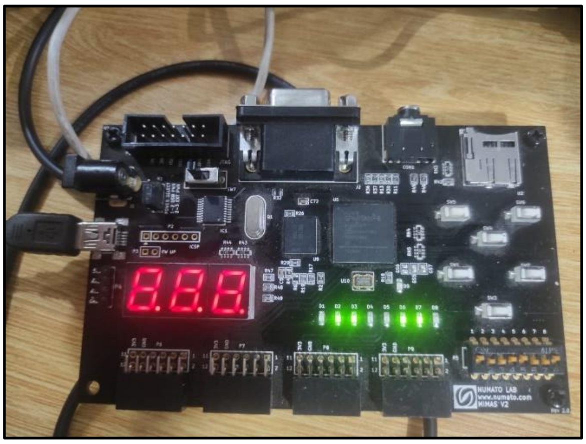
Verilog Code:

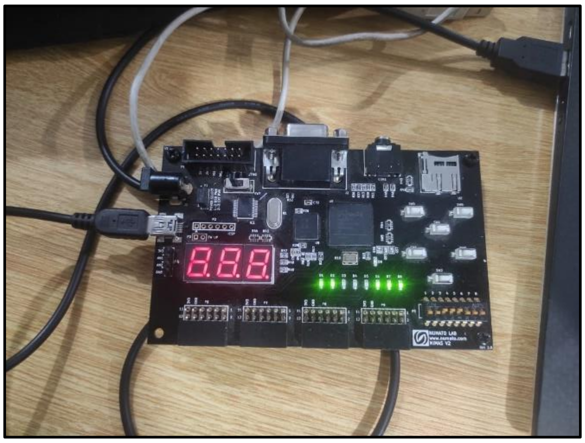
```verilog
module Buf8(
    OUT,IN
     );
     input [3:0] IN;
     output [3:0] OUT;
     assign OUT=IN;


endmodule
```

User constraints File (.ucf):

```
# PlanAhead Generated physical constraints

NET "IN[0]" LOC = F17;
NET "IN[1]" LOC = F18;
NET "IN[2]" LOC = E16;
NET "IN[3]" LOC = E18;
NET "OUT[0]" LOC = P15;
NET "OUT[1]" LOC = P16;
NET "OUT[2]" LOC = N15;
NET "OUT[3]" LOC = N16;

# PlanAhead Generated IO constraints

NET "IN[3]" PULLUP;
NET "IN[2]" PULLUP;
NET "IN[1]" PULLUP;
NET "IN[0]" PULLUP;
NET "OUT[3]" SLEW = FAST;
NET "OUT[2]" SLEW = FAST;
NET "OUT[1]" SLEW = FAST;
NET "OUT[0]" SLEW = FAST;
```

Practical Implementation:

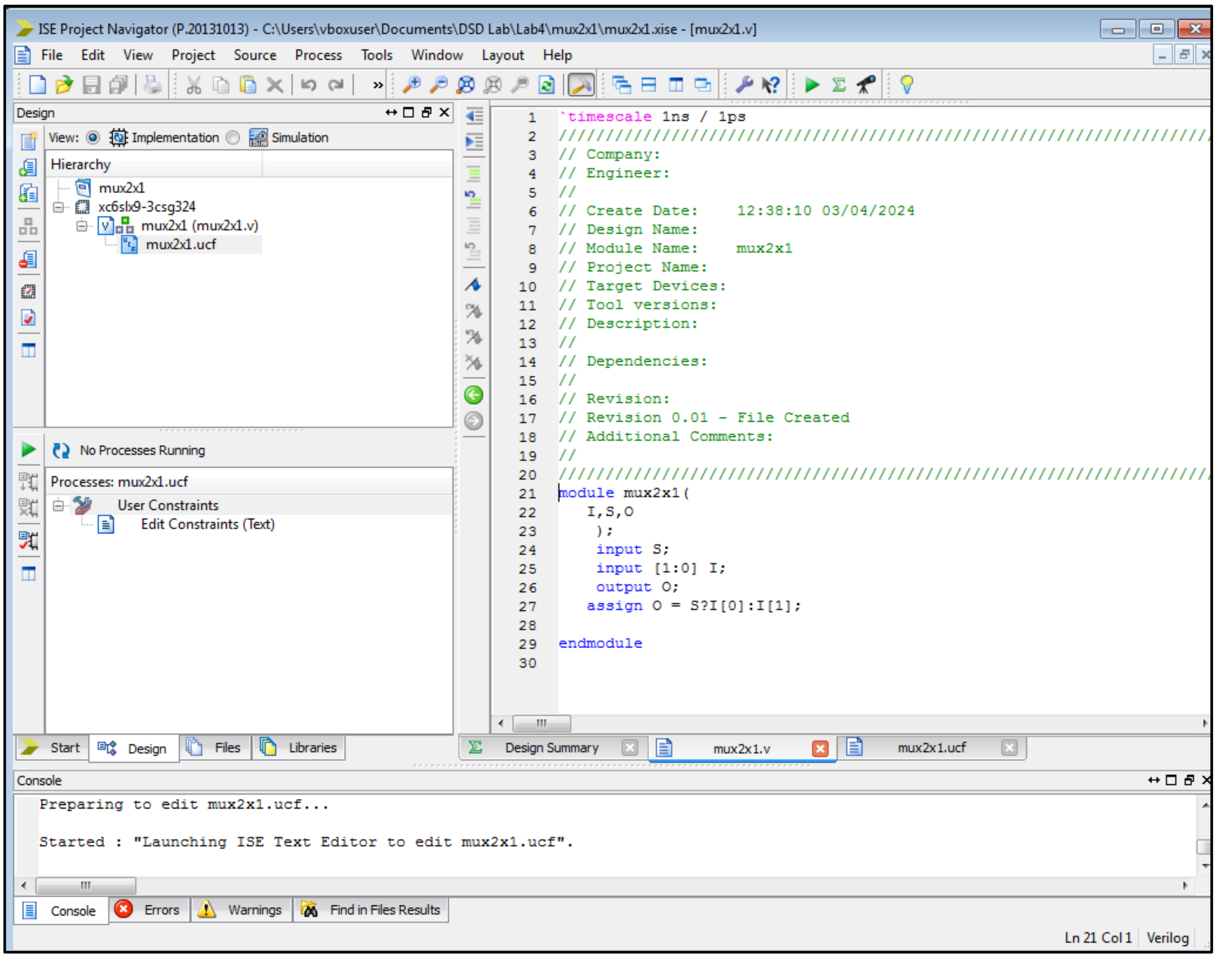




Task 2:

Develop a program that implements a 2x1 multiplexer on the board. Connect the inputs to switches and output to led.

Xilinx ISE Software:



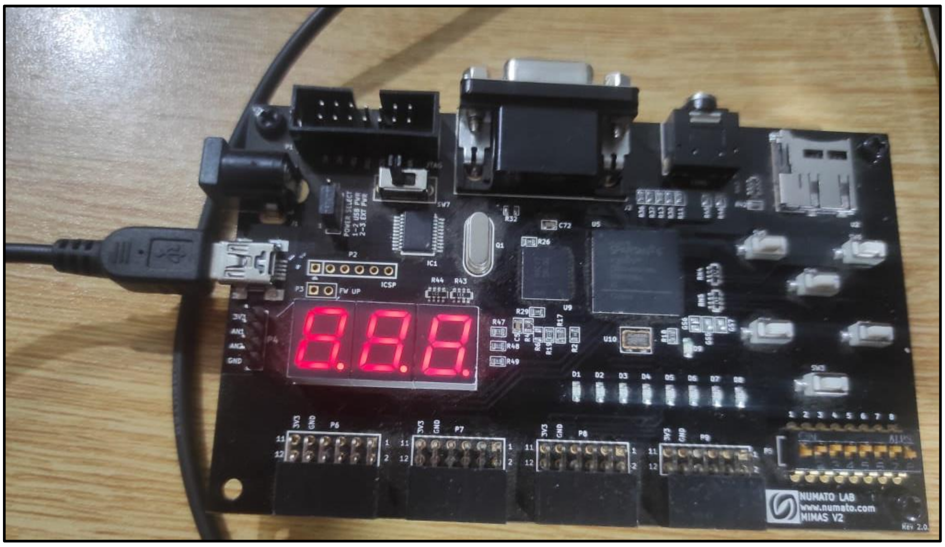
Verilog Code:

```verilog
module mux2x1(
    I,S,O
     );
     input S;
     input [1:0] I;
     output O;
    assign O = S?I[0]:I[1];

endmodule
```

User constraints File (.ucf):

```
# PlanAhead Generated physical constraints

NET "S" LOC = C18;
NET "I[0]" LOC = C17;
NET "I[1]" LOC = D17;
NET "O" LOC = P15;

# PlanAhead Generated IO constraints

NET "S" PULLUP;
NET "I[0]" PULLUP;
NET "I[1]" PULLUP;
NET "O" SLEW = FAST;
```

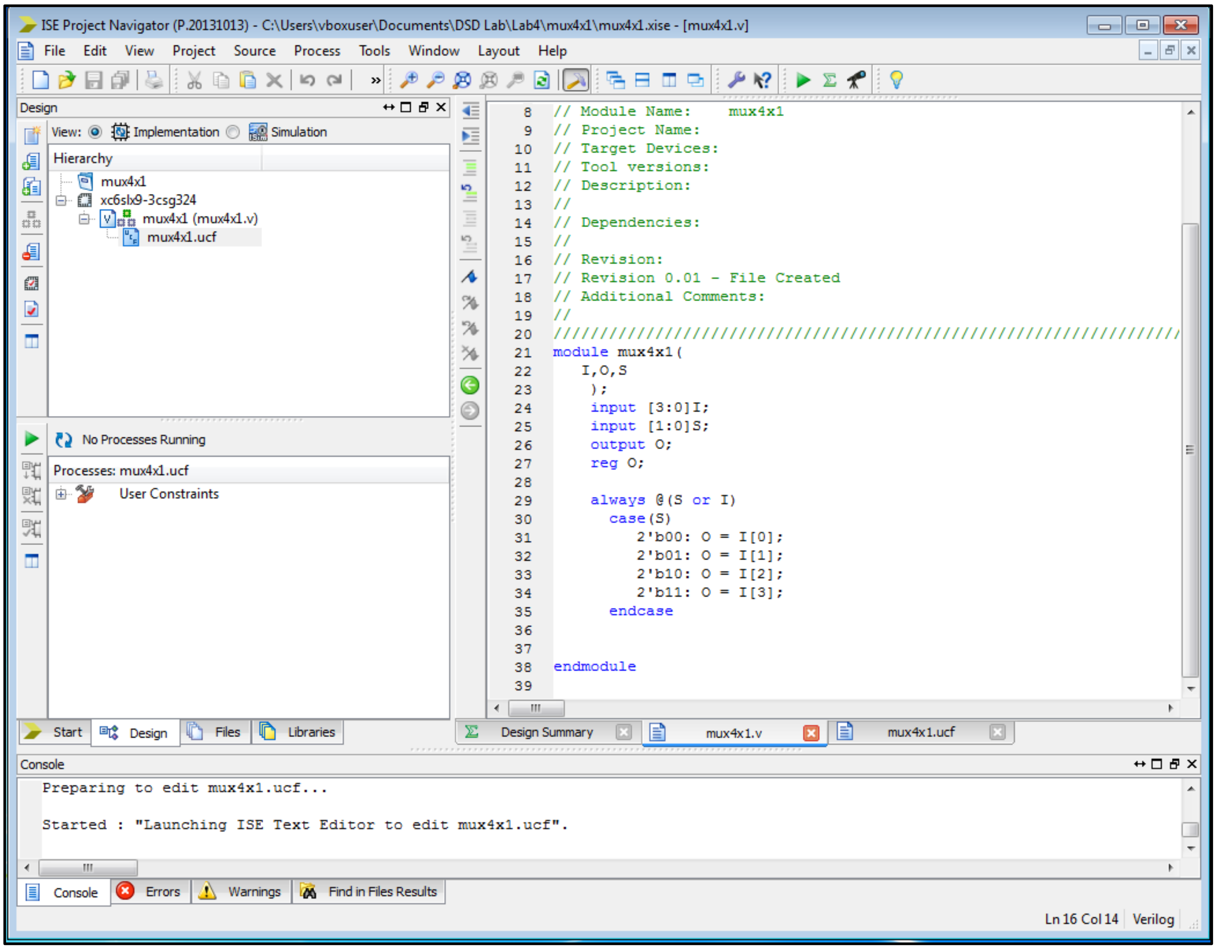
Practical Implementation:



Task 3:

Develop a program that implements a 4x1 multiplexer on the board. Connect the inputs to switches and output to led.

Xilinx ISE Software:



Verilog Code:

```
module mux4x1(
    I,O,S
    );
    input [3:0]I;
    input [1:0]S;
    output O;
    reg O;

    always @(S or I)
      case(S)
         2'b00: O = I[0];
         2'b01: O = I[1];
         2'b10: O = I[2];
         2'b11: O = I[3];
      endcase


endmodule
```

User constraints File (.ucf):

```

# PlanAhead Generated IO constraints

NET "I[3]" PULLUP;
NET "I[2]" PULLUP;
NET "I[1]" PULLUP;
NET "I[0]" PULLUP;
NET "S[1]" PULLUP;
NET "S[0]" PULLUP;
NET "O" SLEW = FAST;

# PlanAhead Generated physical constraints

NET "I[3]" LOC = F17;
NET "I[2]" LOC = F18;
NET "I[1]" LOC = E16;
NET "I[0]" LOC = E18;
NET "S[1]" LOC = D18;
NET "S[0]" LOC = D17;
NET "O" LOC = P15;

```

Practical Implementation: